

The Blind Spot Paradox

Formalizing the race and measuring the adaptation

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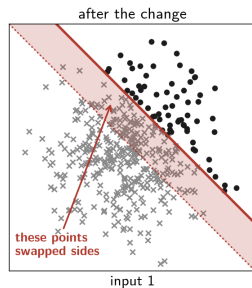
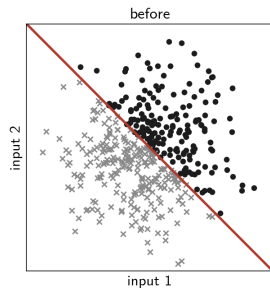
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Concept drift: the rule changes, the data does not

$\underbrace{P(X)}_{\text{the data}}$ unchanged

$\underbrace{P(Y | X)}_{\text{the rule}}$ changes

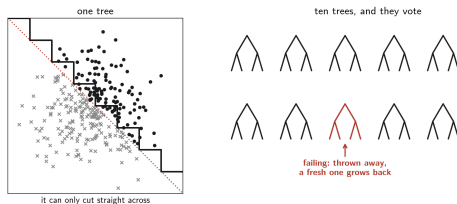


one dot = one observation; its shape is the answer the model has to predict

A model trained before the change is now **wrong on the shaded band**.

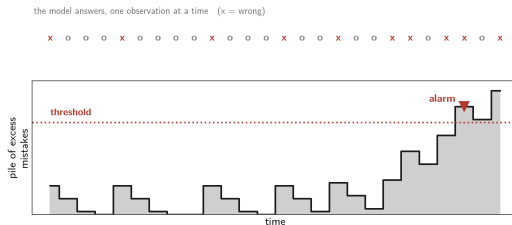
Two defences that run at the same time

The model repairs itself



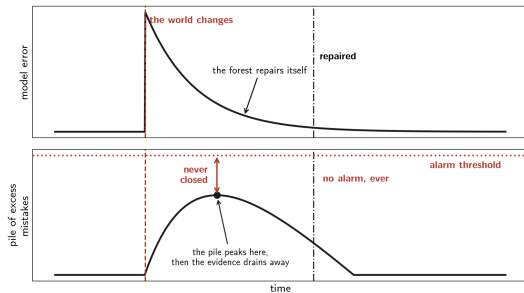
Adaptive Random Forest: ten trees, each replaced when its own error rises.

A monitor raises the alarm



Each error adds to a statistic S_t ; the alarm fires when it crosses λ .

The paradox, and what the reviewers asked



schematic

The paper's claim

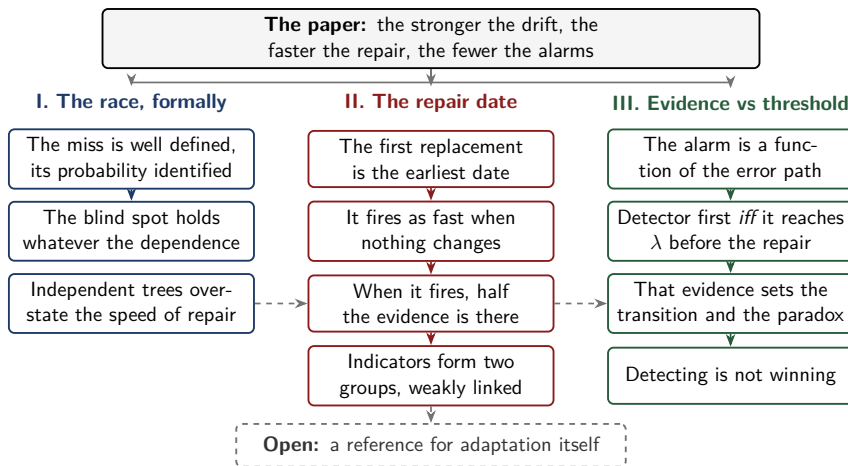
The stronger the drift, the faster the repair, and the **fewer** alarms.

Two corrections requested

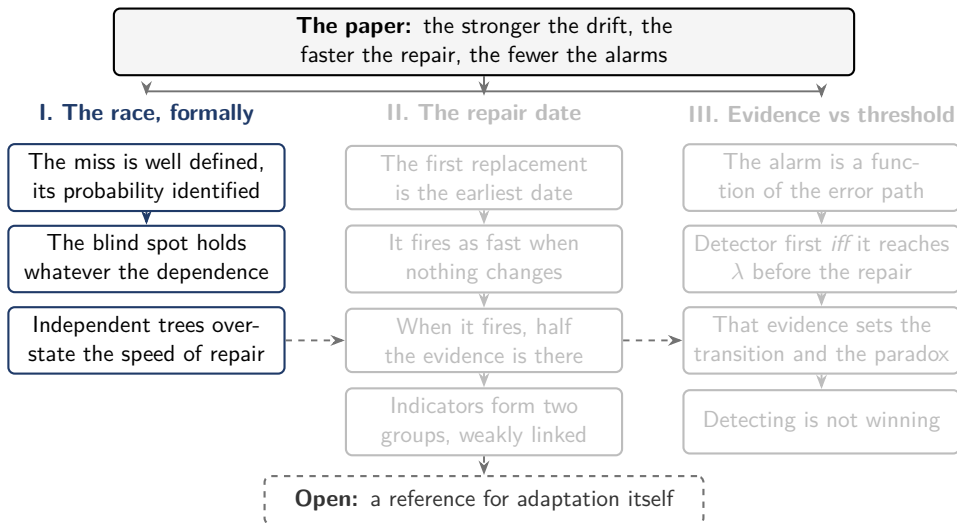
- 1 It races a random repair time against a *constant*, and assumes the trees independent.
- 2 It dates the repair at the *first* tree replaced, without showing that this date means anything.

Our reading, in one map

The blind spot is not a race between two clocks, but a budget of evidence set against a threshold.



Part I — The race, formally



The race is well posed, and its probability identified

One definition, no ad hoc rule

$$\text{Miss} := \{\tau_{\text{ARF}} < \tau_{\text{det}}\}$$

strict inequality; the alarm may never come

- a tie goes to the detector
- a silent repair, with no alarm ever, **is** a miss
- if neither acts, it is not

What the campaign allows

The forest always replaces a tree within the horizon, so no run is left undecided: P_{miss} is **identified**.

all 20 drift sizes pooled:

λ	8	25	50
P_{miss}	0.0905	0.9715	1.0000

Without any dependence assumption

From the marginal laws alone, the blind spot at the weakest drift ($\Delta e = 0.028$, observed 1.00) is at least 0.91 at $\lambda = 8$.

τ_{ARF} : first tree replaced τ_{det} : first alarm Δe : drift size, the jump in error

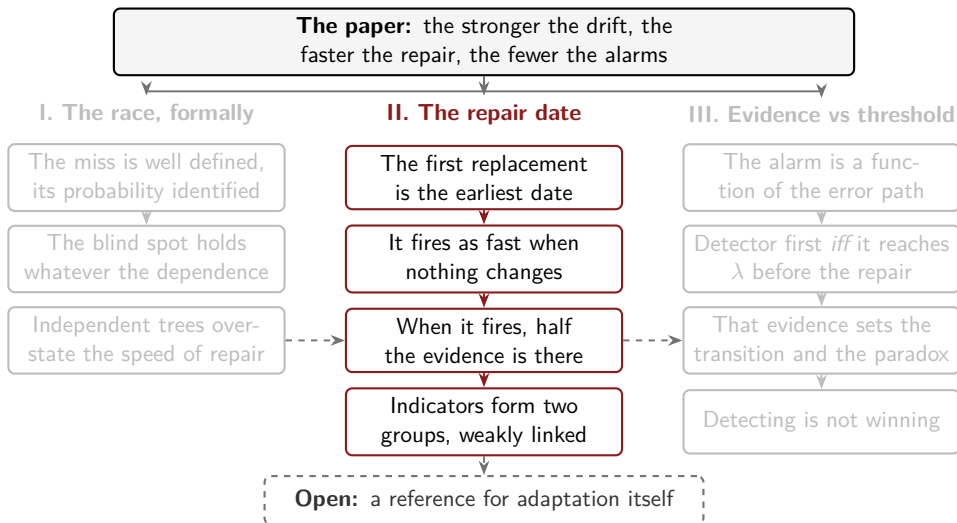
Trees share a stream: the paper is optimistic

- ➊ **Given the stream**, what separates two trees is their own random draws: the trees are independent *conditionally*.
- ➋ **Hence the covariance of two trees is a variance**, never negative: sharing a stream can only make them fail together — the forest is *slower* than the independence formula says.
- ➌ **More precisely**, averaging over streams, convexity gives the bound:

$$P(\tau_{\text{ARF}} \leq s) \leq 1 - (1 - F(s))^M$$

- ➍ **Conclusion:** the independence formula overstates the speed of the repair, hence the blind spot; the paper's critical ensemble size becomes a conservative ceiling, and its verdict does not change.

Part II — What the repair date measures



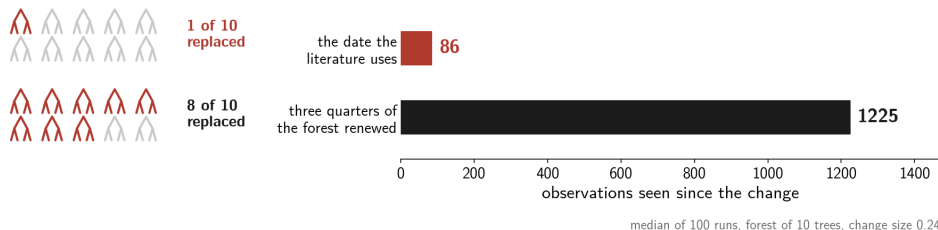
The campaign

- the paper's bench, rerun: 20 drift sizes $\times$ 100 seeds, 2 000 steps after the change
- 2 000 more runs with no change at all
- only the error and the replacement events are recorded; every threshold is replayed offline

Reading an association

- always **within one drift size**: pooling measures the drift size, not the link
- Kendall's rank correlation, with bootstrap intervals over the seeds
- steps added after seeing the data are flagged as exploratory

The first replacement is the earliest possible date



- On every run, the first replacement comes before any other quota of renewed trees. Three quarters of the forest are renewed 6.0 **to** 14.3 **times later**.
- **With no drift at all**, the first replacement comes after 95 steps in median, against 118 on the weakest drift.
- On weak drifts it comes *later*: 346 steps at $\Delta e = 0.085$. With no drift but the same error level, it also waits 268: the error level delays it, not the change.

When the first tree goes, most of the evidence is already there

Evidence at the first replacement

49% to 76%

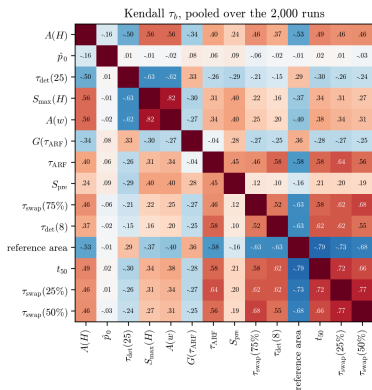
of the detector's final peak, on the 18 drift sizes studied

Does the date track anything, within a drift size?

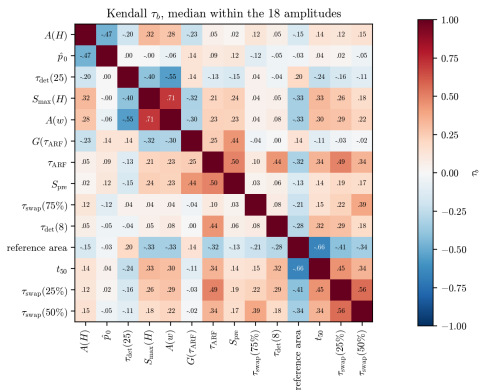
compared with	median link	sizes
detector peak	0.21	15/18
excess error, whole horizon	none	0/18
excess error, transition only	0.23	13/18

The null over the whole horizon is mostly noise: **half** of that area's variance comes from the estimated baseline, multiplied by the horizon.

Unexpected: pooling manufactures agreement



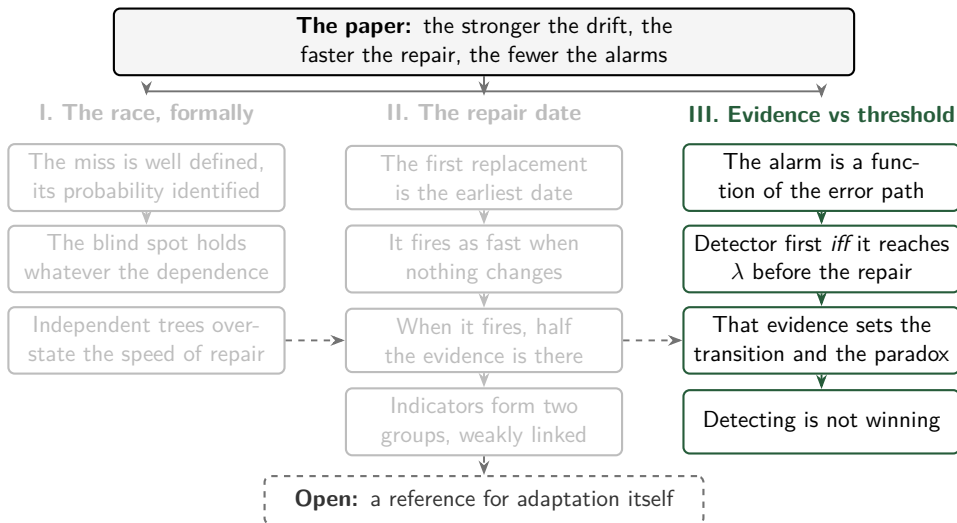
Left, pooled: almost everything agrees, because the drift size moves every indicator.



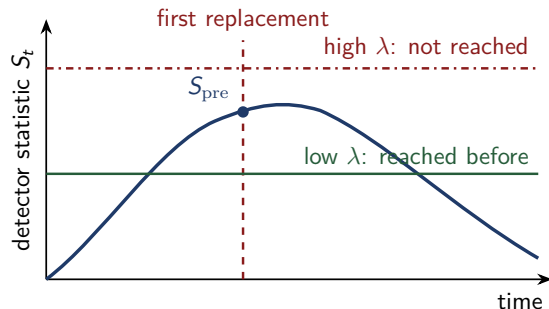
Right, within a size: two groups remain, *evidence* (0.71) and *repair*, weakly joined.

Best date against the analytic reference: **3 trees replaced**, 0.45, ahead of the first tree, 0.34.

Part III — Evidence against the threshold



The race is decided before the repair



schematic

A time counts observations; λ counts excess errors. Comparing them needs a rate that does not exist.

Race identity, on every run

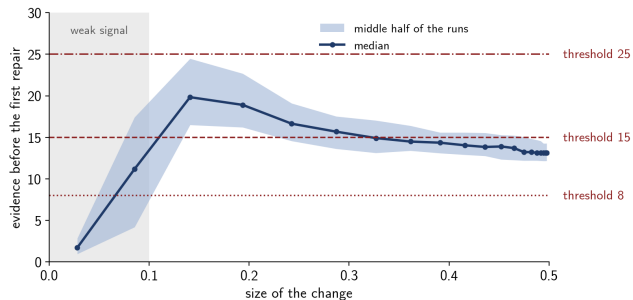
$$\tau_{\text{det}} < \tau_{\text{ARF}} \iff S_{\text{pre}} \geq \lambda$$

$$S_{\text{pre}} = \max_{t < \tau_{\text{ARF}}} S_t$$

S_{pre} : the highest the detector's evidence gets before the forest repairs itself

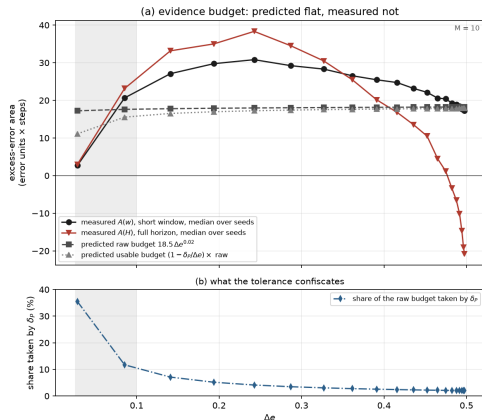
No rate, no averaging, no assumption on the trees. Checked on 2000 runs out of 2000 at three thresholds.

The evidence before the repair sets the transition



- Above the weak-signal zone, the median lies between 13.1 **and** 19.8: the transition the paper sees between 15 and 25, **and leaves unexplained.**
- It **decreases** as the drift grows: at $\lambda = 15$ the detector wins 81%, then only 18%. That is the paradox.

Stronger drifts leave no more evidence

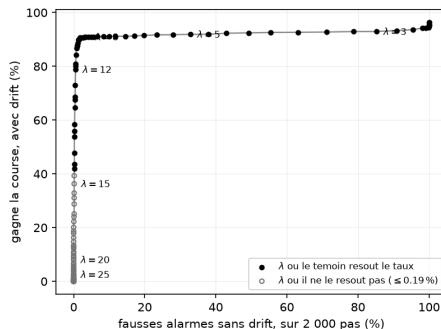


- The paper's picture: an excess error Δe over a window τ_{ARF} , so the area stays almost constant.
- Measured over the transition, the area varies by a factor of 1.8 while the drift size varies by 5.8: **the compensation is real.**
- But the paper's rectangle misestimates it by up to 43%, and over the whole horizon the area turns negative at large drifts: the new task is easier.

Detecting is not winning

λ	alarm	first	false alarms
8	95.3%	91.0%	4.15%
10	94.8%	90.0%	1.30%
12	94.3%	78.8%	0.55%
15	92.9%	36.3%	0.15%
20	66.2%	8.1%	0.00%
25	38.9%	2.5%	0.00%

2 000 runs with drift and 2 000 without, over 2 000 steps



each point is one λ ; the knee sits just below

$$\lambda = 12$$

- At $\lambda = 15$ the alarm almost always fires, but mostly *after* the repair.
- At $\lambda = 10$ the detector still wins 90.0% of races for 1.30% of false alarms: the paper's "no threshold escapes" does not hold on this bench.
- Not tested: the heteroscedastic streams where the paper places its false-alarm tension.

What changes in the paper

The paper	Our result	
paces the repair against a constant	race identified, bounded, decided run by run	made exact
independent trees, bias asserted error back to normal at the first replacement	direction proven, cannot reverse it comes back later; the rectangle fails	proven corrected
first replacement as the repair date	earliest date, fires without drift	qualified
transition between 15 and 25, unexplained	set by the evidence before the repair	explained
no threshold escapes	$\lambda = 8$ and 12 do, on this bench	restricted

The phenomenon survives. What changes is its mechanism and its reach.

An open question, and what we did not do

Open: a reference for adaptation

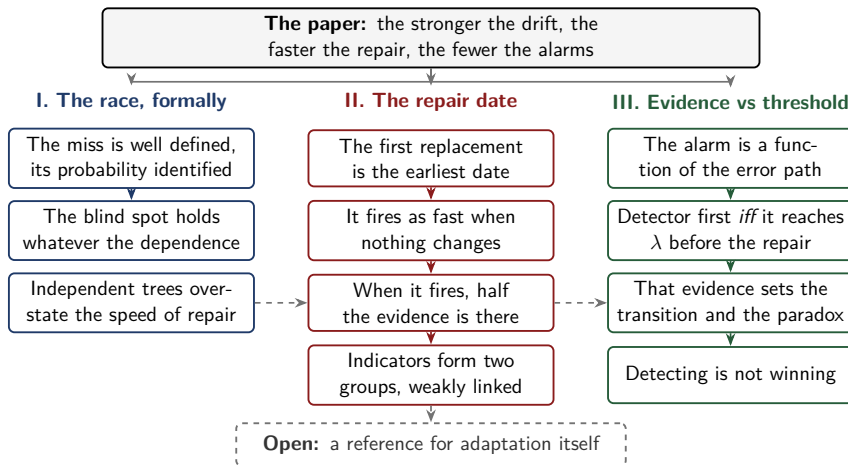
Every indicator on the detector's side reads the *error*, which mixes what the forest relearned with how hard the new task is. On analytic probes, the forest can relearn the changed region and lose a rare one, unseen by the error.

Not done

- one forest size, one homoscedastic bench
- decision rule fixed during the analysis
- the independence bias has a sign, not a size

Take-away

The blind spot is a budget of evidence, accumulated before the repair, set against a threshold.



Backup — The supervisor's additional questions

Treated in a separate working document on the same campaign; not merged into the report.

- **Shape of the error after the drift:** not exponential, and its half-recovery comes *after* the median first replacement on 17 of the 18 readable drift sizes. The first replacement is early, again.
- **End of the horizon:** the error settles *below* its pre-drift baseline at all 20 drift sizes. The new task is easier.
- **Inside the forest** (pilot, 100 runs): disagreement between trees halves when only 63 to 66% of them are renewed — the surviving trees learn too. After a strong drift, new trees err far less than survivors.
- **A z-score of the error drop** measures the number of seeds, not the dynamics.

Open, and worth continuing: a reference for adaptation, the size of the independence bias, heteroscedastic streams, tree-by-tree dynamics.

For integer-valued clocks with marginal CDFs F_A and F_D , and *any* coupling,

$$P_{\text{miss}} \geq \max \left(0, \sup_s [F_A(s) - F_D(s)] \right),$$

$$P_{\text{miss}} \leq \min \left(1, \inf_s [F_A(s) + 1 - F_D(s + 1)] \right).$$

- Lower: Fréchet on $\{\tau_{\text{ARF}} \leq s < \tau_{\text{det}}\} \subseteq \text{Miss}$.
- Upper: $\text{Miss} \subseteq \{\tau_{\text{ARF}} \leq s\} \cup \{\tau_{\text{det}} > s + 1\}$, then Boole.
- Toy case, both clocks uniform on $\{1, 2\}$: the bounds 0 and 0.5 are both attained.

Backup — Lindley's closed form and the certificate

With $X_t = e_t - \hat{p}_0 - \delta_P$ and $A(k, j) = \sum_{t=k+1}^j (e_t - \hat{p}_0)$,

$$S_j = \max_{0 \leq k \leq j} \sum_{t=k+1}^j X_t, \quad S_{\max}(H) = \max_{0 \leq k \leq j \leq H} [A(k, j) - (j - k) \delta_P].$$

- Certificate: the alarm fires by H iff some sub-window carries $A(k, j) \geq \lambda + (j - k) \delta_P$. The full window alone is only sufficient.
- Race identity: the same maximum, taken up to the first replacement, is S_{pre} .

- Every coefficient is computed **within a drift size** (100 seeds), on the 18 sizes with $\Delta e \geq 0.10$: pooling measures the drift size, not the link.
- Kendall's τ_b : reads on pairs, corrects for ties. Censored durations ranked last, licit because the horizon is common to all runs.
- Pearson undefined on censored values; Spearman handles their ties implicitly.
- Percentile bootstrap over seeds, 2 000 resamples; discernible if the interval excludes zero. Fixed during the analysis, not before: stated in the report.
- Complete cases instead of ranking censored runs last: no coefficient moves by more than 0.0403.

Backup — The drift-free control

On 2 000 runs with no change point

runs where at least one tree is replaced	2 000/2 000
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first replacement, median	95 steps
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distinct trees renewed, median	9/10
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forest entirely renewed	28.7%
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first replacement at the weakest drift, median	118 steps
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Δe	drift size: the jump in error a stale model suffers
τ_{ARF}	first tree replaced after the change
$\tau_{\text{swap}}(q)$	first time a share q of the trees has been replaced
τ_{det}	first alarm of the external detector, possibly never
S_t, λ	detector statistic and its threshold
S_{pre}	highest S_t reached before the first replacement
$A(w), A(H)$	excess error over the transient window, over the whole horizon
G	share of the detector's final evidence already reached at τ_{ARF}
